What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the [Università dell'Insubria](https://www.uninsubria.eu/) in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

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Non parametric analysis

A rich zoo of techniques…

- Researchers have developed many methods to cope with the characteristics of light curves in astronomy (irregular sampling, etc.).
- For instance, the Lomb-Scargle (LS) periodogram, epoch folding, analysis of variance (AoV), string length (SL) methods, discrete or slotted autocorrelation, etc.

Lomb Scargle

- As we know already, the LS periodogram is an extension of the conventional periodogram for unevenly sampled time series.
    - A sample power spectrum is obtained by fitting a trigonometric model in a least squares sense over the available randomly sampled data points.
    - The maximum of the LS power spectrum corresponds to the angular frequency whose model best fits the time series.

Epoch folding

- In epoch folding a trial period, P, is used to obtain a phase diagram of the light curve.
  - The trial period is found by an ad-hoc method, or simply corresponds to a sweep among a range of values.
- This transformation is equivalent to dividing the light curve in segments of length P and then plotting the segments one on top of another, hence folding the light curve.
  - If the true period is used to fold the light curve, the periodic shape will be clearly seen in the phase diagram.
  - If a wrong period is used instead, the phase diagram won't show a clear structure and it will look like noise.

Analysis of Variance and String Length

- In AoV the folded light curve is binned and the ratio of the within-bins variance and the between-bins variance is computed.
    - If the light curve is folded using its true period, the AoV statistic is expected to reach a minimum value.
- In SL methods, the light curve is folded using a trial period and the sum of distances between consecutive points (string) in the folded curve is computed.
    - The true period is estimated by minimizing the string length on a range of trial periods.
    - The true period is expected to yield the most ordered folded curve and hence the minimum total distance between points

Discrete autocorrelation

- For the discrete or slotted autocorrelation, time lags are defined as intervals or slots instead of single values.
    - The slotted autocorrelation function at a certain time lag is computed by averaging the cross product between samples whose time differences fall in the given slot.
- All these methods are essentially based on second order statistic analysis.

Information theory based techniques

- Correntropy and mutual information are concepts relatively common in machine learning, and have been applied to time-series analysis with very interesting results.

Phase Dispersion Minimization

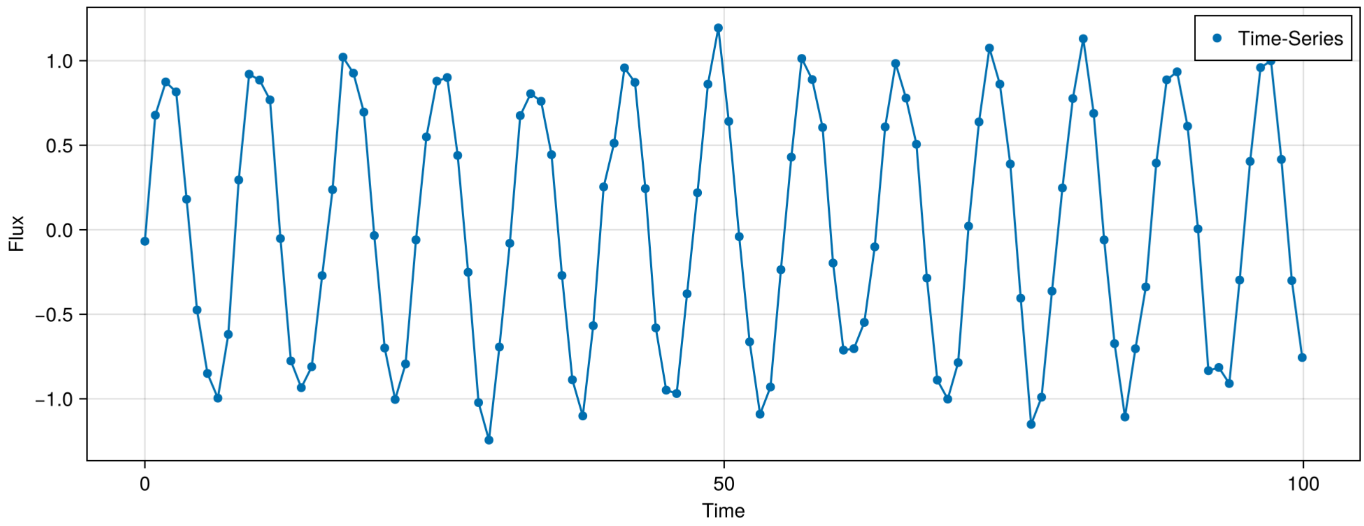
This algorithm was proposed in Stellingwerf (1978).

> This idea consists in distinguishing between possible periods identyfing the one producing the least observational scatter about the mean light curve. Let's see how the algorithm works:

- Let's generate a fake time-series, starting from a sinusoid with a given period and some Gaussian noise. It is instructive to check how the plot looks different without the continuous line joining the points.

```
begin
    PERIOD = 5.

    ts = range(start=0,stop=100,step=0.9)
    fl = sin.(ts .* PERIOD ./ (2*pi)) .+ rand(Normal(0,0.1),length(ts))
end;
```



Mathematical background

- A discrete set of observations can be represented by two vectors, the magnitudes x and the observation times t, where the ith observation is given by (x_i, t_i) and there are N points with $(j = 1, N)$.
- And be σ^2 the sample variance of x, where $\tilde{x}$ is the mean:

$$\sigma^2 = \frac{\sum (x_i - \tilde{x})^2}{N - 1}$$

- Suppose we have chosen M distinct subsamples, having variances s_j^2 $(j = 1, M)$ and containing n_j data points. The overall variance for all the samples is then given by:

$$s^2 = \frac{\sum (n_j - 1)s_j^2}{\sum n_j - M}$$

This can be realized writing, in each of the M subsamples, $s_j^2 = \frac{\sum (x_{ij} - \tilde{x}_j)^2}{n_j - 1}$; then $s^2 = \sum \sum (x_{ij} - \tilde{x}_j)^2 / \sum n_j - M$ and therefore it is clear that s^2 is the variance of our dataset computed as the sum of the variances in each subsample.

Minimizing the variance wrt the mean light curve

- In order to minimize the variance of the data with respect to the mean light curve let Π be a trial period and compute the phases of the input data given the period:

$$\Phi_i = t_j/\Pi - \text{int}(t_j/\Pi)$$

- Now, let's pick M samples from x using the criterion that all the members of sample j have similar ϕ_j.
- The variance of these samples gives a measure of the scatter around the mean light curve defined by the means of the x_j in each sample, considered as a function of ϕ.
- If we define the statistics:

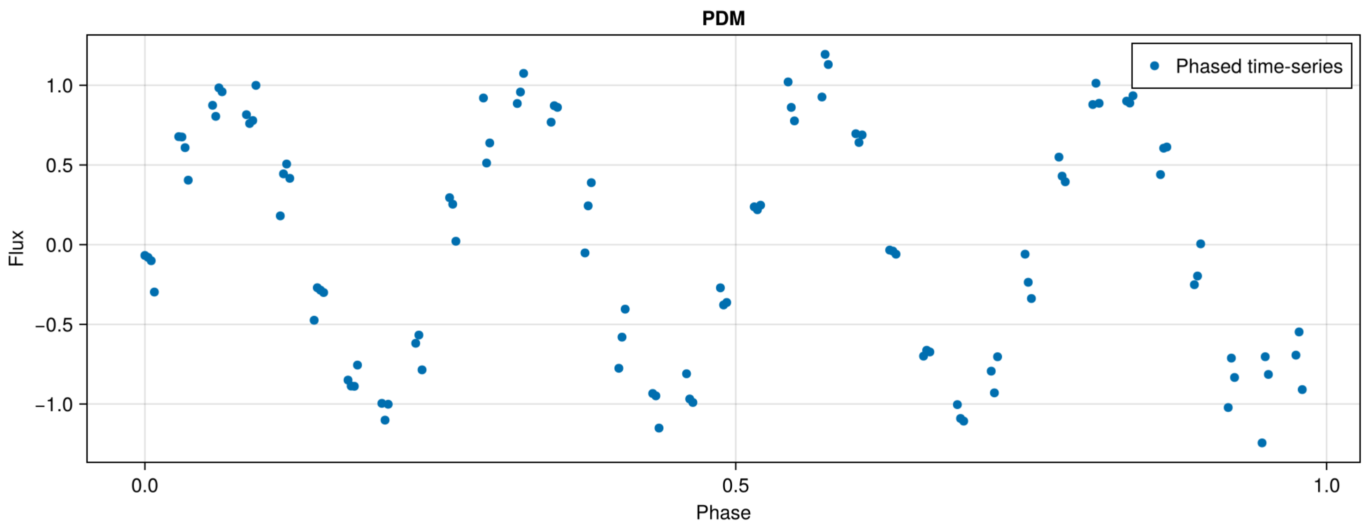
$$\Theta^2 = s^2/\sigma^2$$

- If Π is not the true period, then $s^2 \approx \sigma^2$, while if Π is the true period Θ will be in a local minimum compared to other tested periods.
- Since this technique seeks to minimize the dispersion of the data at constant phase, the author referred to it as "phase dispersion minimization" (PDM).

It can be proven that, mathematically, this algorithm is a least-squares fitting technique, but rather than a fit to a given curve (such as a Fourier component), the fit is relative to the mean curve as defined by the means of each bin.

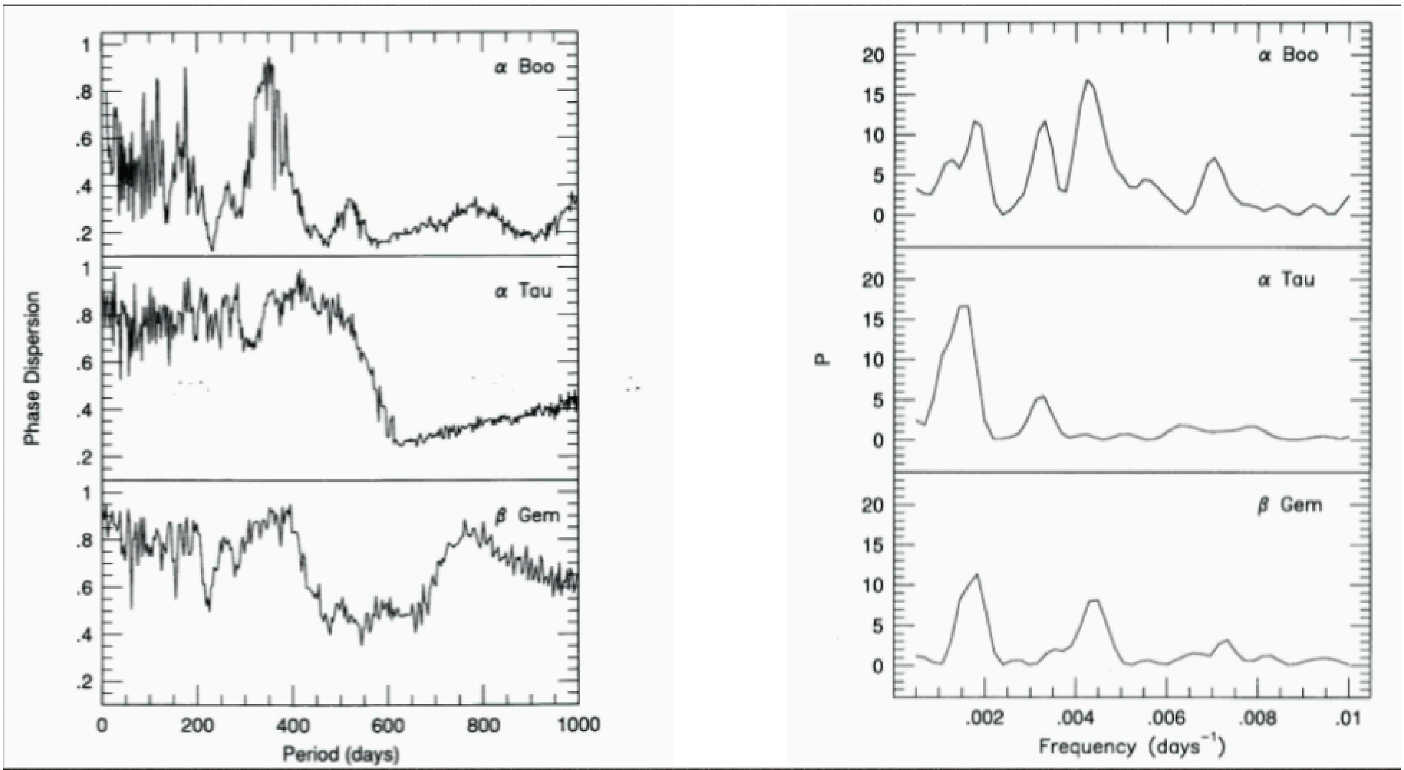
- Let's now try to phase our input data with the right and a wrong period.

Right period? ✅



Statistics of Θ

- To compute Θ, we take the ratio of variances of the two subsets of the input data X, that of the actual observations, and that of the bins.
- Therefore, it was suggested that Θ follows a probability density given by an F distribution with $\sum n_j - M$ and $N—1$ degrees of freedom.
    - Actually, further studies showed this is not correct, and the right statistic turns out to be an incomplete beta distribution although, often, Monte Carlo methods are applied to derive the significance of identified features.
- Several improvements have been proposed since 1978 for this algorithm, to make it more usable for larget datasets or to avoid the possible bias introduced by the (somewhat arbitrary) binning procedure (See, e.g., the wikipedia page devoted to the PDM technique).



String Length Method

- This algorithm was proposed originally in Lafler & Kinman (1965) and than discussed in Dworetsky (1983) and extended in Clarke (2002).

- In this method the quantity to be minimized is the sum of the lengths of line segments joining the observed points reordered according to the phases (m_i, φ_i) in a phase diagram, given a trial period.
- The period is chosen to minimize the following quantity, with N the number of observations:

$$\Theta(P) = \frac{\sum_{i=1}^{N}[(m_{i+1} - m_i)^2]}{\sum_{i=1}^{N}[(m_i - \bar{m})^2]}$$

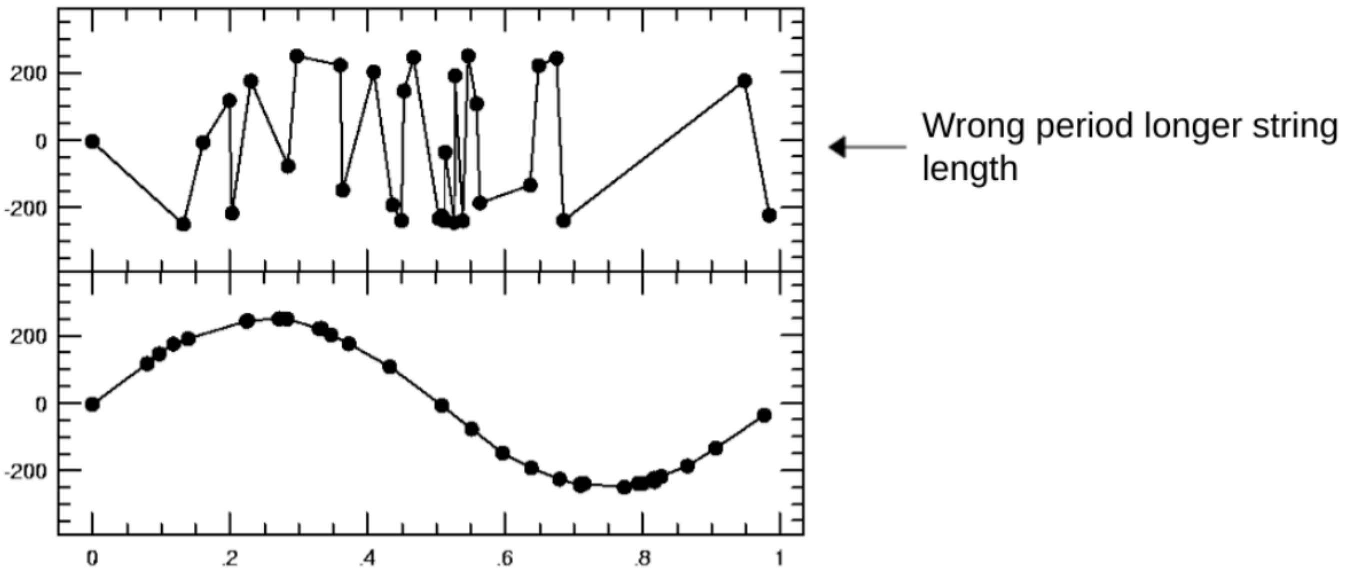
- Please note that it is included the length of the segment joining the last and first measurement in the reordered sequence by letting $m_{N+1} = m_1$.
- By expanding the above formula, and rearranging the indices, we have:

$$\Theta(P) = \frac{2 \times \left(\sum_{i=1}^{N} m_i^2 - \sum_{i=1}^{N}(m_i m_{i+1})\right)}{\sum_{i=1}^{N} m_i^2 - N\bar{m}^2}$$

- The only term depending on the chosen period is $\sum_{i=1}^{N}(m_i m_{i+1})$.
    - If oscillatory behavior is not present, none of the examined periods will provide a correlation and the summation of the products of the phase adjacent measurements is converging to $N\bar{m}^2$.
    - So, any periodogram based on $\Theta(P)$ will fluctuate about a mean value ≈ 2.
- Clarke (2002) introduced a rescaling term $(N-1)/2N$ to remove the sample size bias and normalize the periodogram to 1, leading to the so-calles String-Length-Lafler-Kinman (SSLK) periodogram:

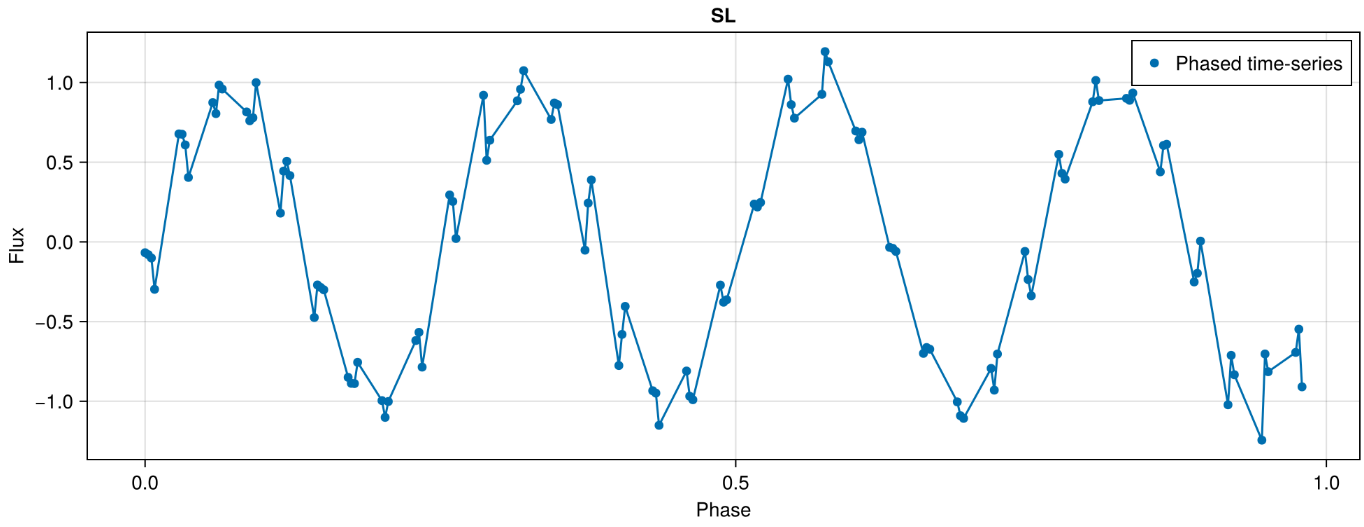
$$T(P) = \frac{\sum_{i=1}^{N}[(m_{i+1} - m_i)^2]}{\sum_{i=1}^{N}[(m_i - \bar{m})^2]} \times \frac{N-1}{2N}$$

- By all means, the algorithm is straightforward:



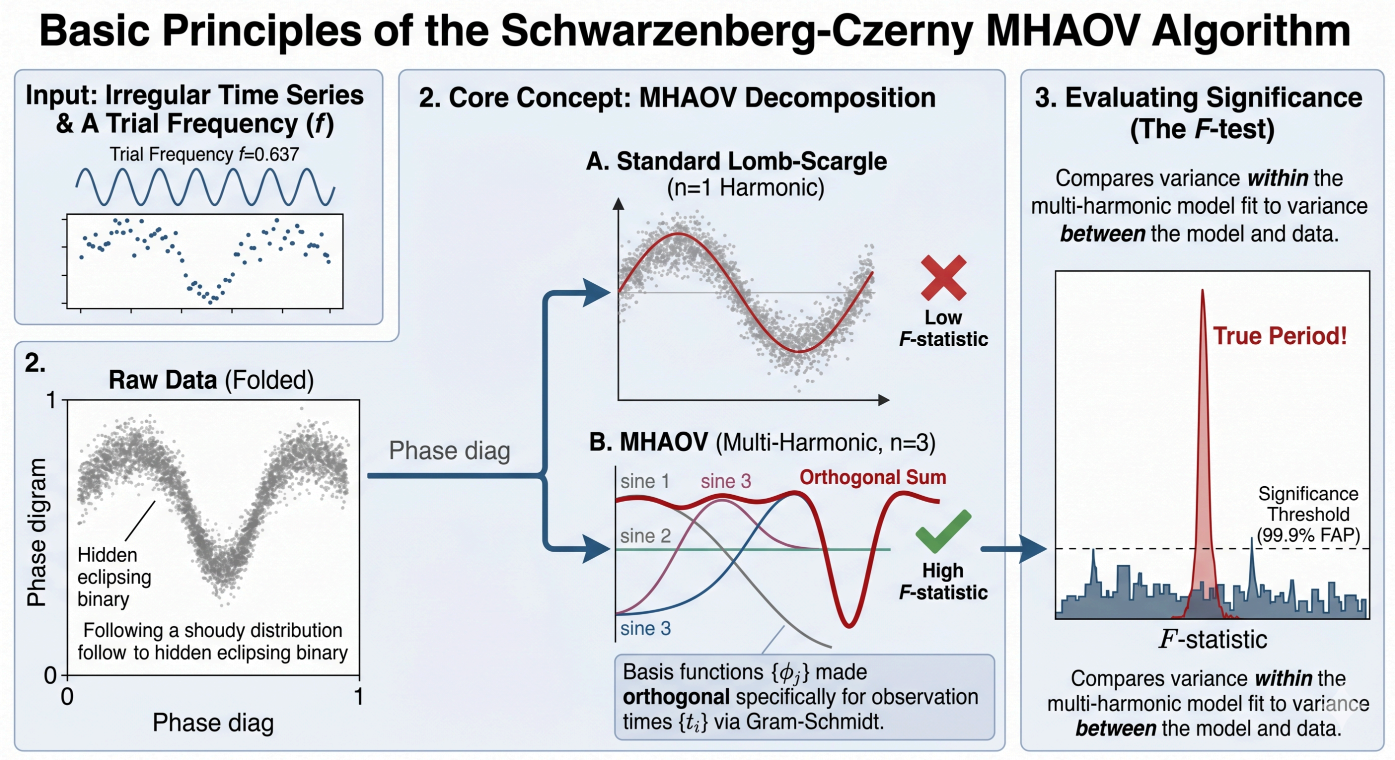
- And again, Monte Carlo methods are the best to determine the singificance of a given period.
- We can see the different path even for the example presenetd above:

Right period? ✅



Analysis of variance and orthogonal polynomials

- This algorithm has been introuced in Schwarzenberg-Czerny (1989) is a phase-folding technique and it treats the search for a period as a statistical grouping problem.
- First, the algorithm picks a “trial period” and “folds” the time-series data into a phase diagram (mapping all data points into a single cycle from 0 to 1). Then, the phase cycle is divided into several “bins” (discrete segments) and, finally, the variance of the data within each bin versus the variance between the bins is computed.
    - If the period is wrong: The data points will be randomly scattered; the variance within each bin will be high, and the means of the bins will be similar.
    - If the period is correct: The data will follow a coherent shape. The points in each bin will be tightly clustered (low internal variance), but the means of the different bins will vary significantly (high between-bin variance).
- This is, indeed, close to the strategy adopted for the PDM technique.
- In Schwarzenberg-Czerny (1996), the same author updated the method to use orthogonal polynomials (often called the MHAOV or Harmonic AoV).
- Instead of just using discrete bins, this version fits the phase-folded data with a series of harmonics making the algorithm not dependent on the, partly arbitrary, choice of the bin length.
- In practise, for the MHAVO algorithm, the period search is a least-squares problem using orthogonal basis functions made by complex polynomials.



▶ Some more mathematical description of the MHAOV algorithm

Generalized Correlation Function: Correntropy

Recently, non-parametric methods have attracted a considerable attention due to novel appoaches, based on ideas and algorithms developed for machine-learning in a big-data scenario, have been developed.

- Huijse et al. (2012) discussed a methodology based on information theoretic (IT) based criteria.
  - IT criteria can extract information from the probability density function, therefore including higher-order statistical moments present in the data.
  - This usually implies a better modelling of the underlying process and robustness to noise and outliers

Huijse et al. (2012) proposed a new information theoretic metric for finding periodicities in stellar light curves.

- The proposed metric combines correntropy (generalized correlation) with a periodic kernel to measure similarity among samples separated by a given period.
- The new metric provides a periodogram, called Correntropy Kernelized Periodogram (CKP), whose peaks are associated with the fundamental frequencies present in the data.
- The CKP does not require any resampling, slotting or folding scheme as it is computed directly from the available samples.
- The generalized correlation function (GCF), i.e., the correntropy, measures similarities between feature vectors separated by a certain time delay in input space.
- The similarities are measured in terms of inner products in a high-dimensional kernel space.
- For a random process $\{X_t, t \in T\}$ with T being an index set, the correntropy function is defined as:

$$V(t_1, t_2) = \mathbb{E}_{x_{t_1} x_{t_2}}[\kappa(x_{t_1}, x_{t_2})]$$

- It is also possible to define the centered correntropy:

$$U(t_1, t_2) = \mathbb{E}_{x_{t_1} x_{t_2}}[\kappa(x_{t_1}, x_{t_2})] - \mathbb{E}_{x_{t_1}} \mathbb{E}_{x_{t_2}}[\kappa(x_{t_1}, x_{t_2})]$$

- where $\mathbb{E}[\cdot]$ denotes the expectation value and $\kappa(\cdot, \cdot)$ is any positive definite kernel.
- A kernel can be viewed as a measure of the similarity of the data. A frequently used kernel is the Gaussian kernel:

$$G_\sigma(x - z) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{\|x - z\|^2}{2\sigma^2}\right)$$

- The kernel size gives the user the ability to control the emphasis given to the higher-order moments with respect to second-order moments.
  - For large values of the kernel size, the second-order moments have more relevance and the correntropy function approximates the conventional correlation.
- For a discrete strictly stationary random process $\{X_n\}$, the univariate correntropy function or autocorrentropy can be defined as:

$$V[m] = \mathbb{E}[\kappa(x_n, x_{n-m})]$$

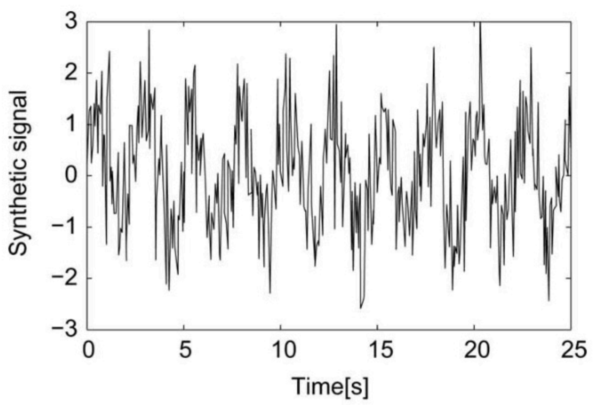
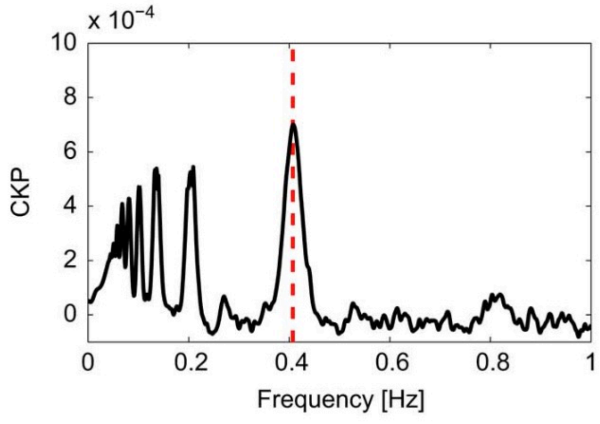
- Correntropy outperformed conventional correlation, showing better discriminatory and robustness to noise.

- Kernels can also be viewed as covariance functions for correlated observations at different points of the input domain (see lectures about Gaussian Processes).
    - A kernel can be any semi-definite positive function. And it is possible to develop periodic kernels.

- For instance, one possibility is:

$$G_{\sigma;P}(z-y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\sin^2\left(\frac{\pi}{P}(z-y)\right)}{0.5\sigma^2}\right)$$

- where now the kernel depends explictly on the period.
- This brings to a metric combining the correntropy with a periodic kernel to measure similarity among samples separated by a given period.
- This algorithm is known as Correntropy Kernelized Periodogram, and does not require any resampling, slotting or folding scheme, as it is computed directly from the available samples.
- This idea has strong connection with Gaussian Process analysis, machine learning and information theory.

Simulated data	CKP Periodogram
	

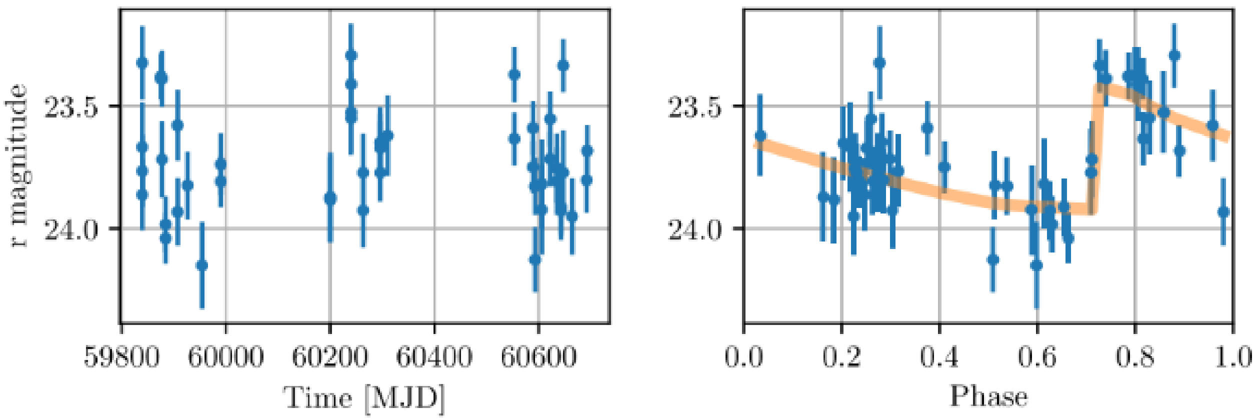
- Again, Monte Carlo methods are usually applied to estimate the significance of the identified periodogram peaks.

Mutual Information

> Another modern approach to the problem tries to measure the degree of dependence between different random variables (RVs). MI is able to capture nonlinear dependence too.

This idea was introduced Huijse et al. (2018), where a new method for light curve period estimation based on quadratic mutual information (QMI) is discussed.

- The proposed method does not assume a particular model for the light curve nor its underlying probability density and it is robust to non-Gaussian noise and outliers.
- MI measures the reduction of the uncertainty of a random variable (RV) given that we know a second RV.
    - MI can also be seen as a measure of dependence although, unlike correlation, MI is able to capture nonlinear dependence between RVs.
- Technically speaking, the MI is the divergence (i.e. statistical distance) between the joint PDF of the RVs and the product of their marginal PDFs.
- For the interested readers, an exhaustive description of the algorithm is reported here (notebook, html).
- If the light curve is folded with the wrong period, the structure in the joint PDF will be almost equal to the product of the marginal PDFs, i.e., magnitudes are independent of the phases.
- On the other hand, if the correct period is chosen, the joint PDF will present a structure that is not captured by the product of the marginals. By maximizing MI we are maximizing the dependency between model and observations.



Reference & Material

Material and papers related to the topics discussed in this lecture.

- Clarke (2002) - "String/Rope length methods using the Lafler-Kinman statistic"
- Dworetsky (1983) - A period-finding method for sparse randomly spaced observations or “How long is a piece of string?”
- Huijse et al. (2018) - Robust Period Estimation Using Mutual Information for Multiband Light Curves in the Synoptic Survey
- Huijse et al. (2012) - An Information Theoretic Algorithm for Finding Periodicities in Stellar Light Curves
- Schwarzenberg-Czerny (1996) - "Fast and Statistically Optimal Period Search in Uneven Sampled Observations"
- Stellingwerf (1978) - Period Determination Using Phase Dispersion Minimization

Further Material

Papers for examining more closely some of the discussed topics.

- Lafler & Kinman (1965) - "An RR Lyrae Star Survey with Ihe Lick 20-INCH Astrograph II. The Calculation of RR Lyrae Periods by Electronic Computer"
- Krishnan et al. (2021) - Detection of periodic signals in AGN red noise light curves: empirical tests on the Auto-Correlation Function and Phase Dispersion Minimization
- Schwarzenberg-Czerny (1997) - The Correct Probability Distribution for the Phase Dispersion Minimization Periodogram

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Science case about FRBs	Lecture about non-parametric periodograms	Course Summary
html	Science case about FRBs	Lecture about non-parametric periodograms	Course Summary

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